AOU
الجامعة العربية المفتوحة
Arab Open University
Faculty of Computer Studies

MS102
Physics

Final Examination
Spring Semester 2025/2026

Date: 08/June/2026

Number of Exam Pages: 4 (including this cover sheet)	**Time Allowed:** 2hrs

Instructions:

- Total Marks: **50**
- This exam consists of **2** parts.
- **ALL questions** must be answered in the External Answer booklet.
- Be sure you write your **name and ID** on the External Answer booklet.
- **Calculators** are allowed.

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PART I: Answer the entire Multiple-Choice Questions (MCQ). [25 Marks]

1. 1200 kg car travels at 3.0 m/s and strikes a wall. The car bounces off the wall with a velocity of 4.0 m/s in the opposite direction. If the car is in contact with the wall for 0.10 seconds, then the magnitude of the average force of the wall on the car is
a) 46,000 N b) 84,000 N c) 63,000 N d) 75,000 N

2. The units of momentum are
a) ML/T b) M/T c) L/T^2 d) ML/T^2

3. Which type of wave does not require a medium to propagate?
a) Longitudinal wave b) Transverse wave
c) Mechanical wave d) Electromagnetic wave

4. The rate of change of displacement with respect to time is called:
a) Velocity b) Speed c) Acceleration d) Distance

5. An object is moving with constant acceleration. Which of the following statements is true?
a) Its velocity is constant b) Its acceleration is zero
c) Its velocity is changing at a constant rate d) Its displacement is zero

6. Sound waves are an example of which type of wave?
a) Transverse wave b) Electromagnetic wave
c) Mechanical wave d) Longitudinal wave

7. The intensity of a wave is related to its:
a) Square of Amplitude b) Frequency
c) Wavelength d) Speed

8. Impulse is defined as the:
a) Change in momentum b) Force applied over a period
c) Product of mass and acceleration d) Rate of change of velocity

9. The SI unit of force is:
a) Newton b) Kilogram c) Meter d) Second

10. Which of the following is an example of a contact force?
a) Gravitational force b) Magnetic force
c) Frictional force d) Electrostatic force

11. The speed of a wave in a string is directly proportional to the:
a) Density of the string b) Tension in the string
c) Length of the string d) Wave frequency

12. The distance between two consecutive identical points on a wave is known as:
a) Amplitude b) Frequency c) Wavelength d) Period

13. If the speed of the ball is doubled without affecting its mass, then the ball's momentum will be
a) 2 times larger. B) 2 times larger. C) 3 times larger. d) Not change.

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14. The force required to accelerate an object of mass 4 kg at a rate of 4 m/s^2 is:
a) 16 N b) 1 N c) 8 N d) 0 N

15. The oscillations of particles in a longitudinal wave occur:
a) Perpendicular to the direction of wave propagation
b) In the same direction as the wave motion
c) At right angles to the wave amplitude
d) In random directions

16. A wave passes through a medium at a frequency of 20 Hz. What is the period of the wave?
a) 20 seconds b) 0.05 seconds c) 0.2 seconds d) 2 seconds

17. What is the distance between adjacent nodes in a standing wave?
a) wavelength/4 b) wavelength/3 c) wavelength/2 d) 2 wavelengths

18. When two positively charged objects are brought close together, they will:
a) Attract each other b) Repel each other
c) Cancel each other's charge d) Have no effect on each other

19. Coulomb's law describes the relationship between electric force, charges, and:
a) Time b) Distance c) Speed d) Energy

20. If the distance between two-point charges is doubled, the force between them will:
a) Quadruple b) Half c) Double d) Remain unchanged

21. The opposition offered by a material to the flow of electric current is called:
a) Conductivity b) Voltage c) Resistance d) Current

22. Kirchhoff's first law is related to:
a) Conservation of charge b) Conservation of energy
c) Conservation of momentum d) Conservation of voltage

23. What property of an object determines whether it gains or loses electrons?
a) Mass b) Volume c) Electric charge d) Electric field strength

24. The resistance of a wire depends on its:
a) Length only b) Cross-sectional area only
c) Material and length d) All the above

25. Four 12 Ω resistors are connected. What is the least resistance that can be attained with these resistors by connecting them in various ways?
a) 12 Ω b) 6.0 Ω c) 3.0 Ω d) 2.0 Ω

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Part II: Answer the following questions. [25 Marks]

Question 1: (15 Marks)

a) A string is stretched with a tension of 120 N. The string has a mass of 10.0 grams and has a length of 2.50 meters. Calculate the velocity of wave propagation along the string.(5 Marks)

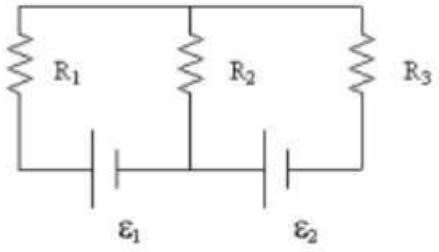
b) Four point charges are located at the corners of a square, 1.00 m by 1.00 m. On each of two diagonally opposite corners are 1.00 μC charges. On each of the other two corners are −1.00 μC charges. What is the magnitude of the force on one of the positive charges? (10 Marks)

Question 2: (10 Marks)

In the diagram, if the current through R_1 is 5 A up, and the current through R_3 is 3 A up,

a) What is the current in R_2? (15 marks)

b) If R1=4Ω, R2=2 Ω, R3=4 Ω What is the equivalent resistance of circuit? (10 Marks)



End of questions

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